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Xiaoai Jiedu recipe reduces cell survival and induces apoptosis in hepatocellular carcinoma by stimulating autophagy via the AKT/mTOR pathway

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A B S T R A C T

Ethnopharmacological relevance: The Xiaoai Jiedu recipe (XJR) is a traditional Chinese medicine formulation used in clinical settings to treat liver cancer. It has shown promising effectiveness by combining herbal and animal-derived ingredients, offering a new approach to cancer treatment. However, its mechanism of action is poorly understood.

Aim of the study: The molecular processes underlying the inhibitory effects of the XJR on hepatocellular cancer (HCC) were investigated.

Materials and methods: The primary chemical components of XJR and associated disease targets relevant to HCC were anticipated and compiled using a database. The potential targets and processes by which XJR influenced HCC were investigated using GO and KEGG enrichment analyses, as well as protein-protein interaction (PPI) networks. Transmission electron microscopy, laser confocal microscopy, and Western blotting were used to evaluate autophagy, while CCK-8 assays measured cell viability and Western blotting and flow cytometry evaluated apoptosis. *In vivo* assays were conducted employing an HCC xenograft mouse model.

Results: Network pharmacology analysis identified 456 intersecting targets between XJR and HCC. The top five active components are quercetin, cholesterol, jaceosidine, eupafolin, and oleanolic acid. The key targets include TP53, AKT1, IL6, EGFR, SRC, HSP90AA1, TNF, IL1B, MYC, and CASP3. Additionally, the autophagy pathway was found to be one of the main pathways through which XJR intervenes in HCC. The increased quantity of autophagosomes and autolysosomes, the overexpression of Beclin1 and LC3A/B-II proteins, and the down-regulation of P62 all suggest that XJR stimulated autophagy in HCC cells. Functional tests employing pathway-specific activators and inhibitors and siRNA-based knockdown demonstrated that XJR promoted autophagy by blocking AKT/mTOR signaling. Furthermore, XJR reduced the viability of HCC cells and promoted apoptosis by upregulating apoptosis proteins. Autophagy inhibitors and Beclin1 silencing reversed these effects. Research conducted *in vivo* showed that XJR activated autophagy through the AKT/mTOR axis, thereby markedly reducing tumor growth and inducing tumor cell demise.

Conclusions: These studies show that XJR activates autophagy in both cellular and animal models to induce apoptosis and decrease HCC cell proliferation, as shown by network pharmacology and verification assays.

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Further, these findings provide experimental evidence that the anti-tumor activity of XJR involves autophagy stimulation.

1. Introduction

Liver cancers are both common and highly aggressive. As indicated by the Global Cancer Statistics of 2022, liver cancer ranks third in cancer-related deaths (Bray et al., 2024). About 90% of liver cancers are hepatocellular carcinomas (HCCs) (Llovet et al., 2021). There has been much progress during the past decade in treating HCC. Treatment techniques that improve survival include liver transplantation, chemotherapy, radiation, immunotherapy, surgical resection and ablation, and molecular targeted drugs. However, current treatments are limited by long incubation periods, multidrug resistance, disease recurrence, and the high incidence of drug-related adverse effects (Forner et al., 2019; Zhang et al., 2021). Thus, effective medications and novel therapies for HCC are urgently needed.

Autophagy is an intricate and highly preserved intracellular process by which senescent and malfunctioning organelles are targeted by double-membrane autophagosomes and delivered to lysosomes for degradation (Cao et al., 2017). Cellular self-digestion maintains cellular homeostasis and survival (Fan et al., 2020). autophagy and apoptosis are categorized as programmed cell death types I and II (X. Li et al., 2020). Autophagy either enhances or antagonizes the apoptosis effect (Xie et al., 2020). Furthermore, autophagy either promotes or inhibits tumorigenesis and cancer progression (Ferrari et al., 2022). Because autophagy plays a complex function in cellular destiny, it is a prospective target for cancer therapy (X. Li et al., 2020).

Traditional Chinese medicine (TCM) has been employed for over three millennia to prevent and treat various cancers, such as HCC, demonstrating significant therapeutic efficacy and minimal harm to normal cells (Huang et al., 2021; Luo et al., 2019; Xiang et al., 2019). Xiaoai Jiedu recipe (XJR), a TCM formulation developed by Professor Zhou Zhongying, contains seven Chinese herbs (Table 1). XJR reduces lung cancer progression by promoting apoptosis via the MAPK pathway (Wang et al., 2019) and decreases HCC proliferation via the miR-200b-3p/Notch1 axis (Qiu et al., 2020). In contrast to other drugs whose efficacy was demonstrated only in experimental studies, XJR substantially improves short-term clinical outcomes in advanced cancers (Zhou et al., 2010). Furthermore, as TCM targets a wide variety of factors and pathways, XJR cures cancer and decreases the toxicity of chemotherapy drugs, greatly enhancing patient quality of life (Zhou et al., 2010). Consequently, a better understanding of the anti-tumor activity of XJR can improve the clinical use of this formulation. Therefore, the present research examined the molecular processes that underlie XJR's impact on HCC, particularly on autophagy.

2. Study methodology

2.1. Materials

The XJR was provided by Anhui Bozhou Traditional Chinese and Western Medicine Co., Ltd. and Suzhou Tianling Traditional Chinese Medicine Drinking Tablets Co. (Anhui, China). A pharmacology professor from Nanjing University of Chinese Medicine authenticated these materials following the Chinese Pharmacopoeia to confirm their authenticity, it was produced by our hospital's preparation room following standardized procedures, with uniform testing standards. We also published a paper with analytical data on the herbal drug investigate(H et al., 2020). Every plant name has been verified using http://www.worldfloraonline.org, and the website was consulted on October 3, 2024, as stated. The decoction was concentrated to 2 g/mL under vacuum and sterilized by filtration. The following kits and reagents were used: BCA Protein Quantification Kit (Beyotime; Shanghai, China); RPMI 1640 medium, DMEM medium, and trypsin (Thermo Fisher Scientific; Waltham, USA); MK-2206 (MK), hydroxychloroquine sulfate (HCQ), fetal bovine serum (Biological Industries, Kibbutz Beit-Haemek, Israel); Annexin V-FITC Cell Apoptosis Detection Kit, Cell Counting Kit-8 (CCK-8) (Dojindo, Kumamoto, Japan); RIPA lysis buffer, and BCA Protein Quantification Kit (Sigma-Aldrich; St. Louis, MO, USA); and SC79 (SC) (Selleck Chemical; Houston, TX, USA). Spreading Hedyotis Herb (Lot No.: 2021082201), Appendiculate Cremastra Pseudobulb (Lot No.: CG08001), Fruit of Fiverleaf Akebia (Lot No.: CR02801), Stiff Silkworm (Lot No.: 2022033103), Centipede (Lot No.: CD03101), Heterophylly Falsestarwort Root (Lot No.: 2022022301), Dwarf Lilyturf Root Tuber (Lot No.: CG01601)

2.2. Network pharmacology

A component and target search for the herbs *Herba Hedyotidis, Cremastrae Pseudobulbus Pleiones Pseudobulbus, Pseudostellariae Radix, and Fructus Akebiae* in the Xiaoai Jiedu formula was conducted using the TCM System Pharmacology (TCMSP, http://ibts.hkbu.edu.hk/LSP/tcmsp.php) database. The screening criteria established were oral bioavailability (OB) $\geq$ 30% and drug-likeness (DL) $\geq$ 0.18. For herbal constituents absent from the TCMSP database, the Herbal Ingredients' Targets (HIT 2.0) platform (HIT-index: badd-cao.net) was utilized to gather active compounds. The SwissTargetPrediction (http://www.swisstargetprediction.ch/) database projected possible targets. UniProt (https://www.uniprot.org/) was used for standardizing target names. Searching for "Hepatocellular carcinoma," the GeneCards database (https://www.genecards.org/) gathered disease-related genes using a relevance score threshold set at >2.9. Herbal-compound-target network was built using Cytoscape 3.9.1. With a minimum needed interaction

Table 1
Formulation of Xiaoai Jiedu recipe (XJR).

Pinyin name	English name	Latin Pharmaceutical Name	Doses	Origin
Baihuasheshecao	Spreading Hedyotis Herb	Herba Hedyotidis	20g	The dried whole her of *Scleromitrion diffusum* (Willd.) R.J.Wang.
Shancigu	Appendiculate Cremastra Pseudobulb	Cremastrae Pseudobulbus Pleiones Pseudobulbus	10g	The dried pseudobulb of *Cremastra appendiculata* (D. Don) Makino.
Bayuezha	Fruit of Fiverleaf Akebia	Fructus Akebiae	12g	The Dried fruit of *Akebia quinata* (Thunb. ex Houtt.) Decne.
Jiangcan	Stiff Silkworm	Bombyx Batryticatus	10g	The dried body of the larvae of *Bombyx mori* L that has been killed due to infection (or inoculation) of the fungus Beauveria bassiana (Bals.) Vuill.
Wugong	Centipede	Scolopendra	3g	The dried body of *Scolopendra subspinipes* mutilans L. Koch
Taizisheng	Heterophylly Falsestarwort Root	Pseudostellariae Radix	15g	The dried tuberous root of *Pseudostellaria heterophylla* (Miq.) Pax et Hoffm.
Maidong	Dwarf Lilyturf Root Tuber	Ophiopogonis Radix	12g	The dried tuberous root of *Ophiopogon japonicas* (Thunb.) Ker-Gawl.

score of>0.7, a PPI network for the intersecting genes was constructed using STRING (https://string-db.org/). Then Cytoscape 3.9.1 assisted in visualizing the PPI network. Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment studies and Gene Ontology (GO) on the common targets were conducted using R's clusterProfiler package (Wu et al., 2021).

2.3. Serum preparation

Qing Long Shan Dong Wu Fan Zhi Chang (Nanjing, China) provided an average of sixteen white female rabbits weighing 2.15 ± 0.26 kg. Based on the XJR dose, the animals were split into four groups at random: low (XJR-L, 1.89 g/kg), moderate (XJR-M, 3.78 g/kg), and high (XJR-H, 7.56 g/kg) doses, and control. The intervention groups received the corresponding dose of XJR intragastrically once daily for 5 days. The control group was administered physiological saline via intragastric route. Blood samples were obtained through the carotid artery 2 h after the final administration of XJR. Centrifugation was used for collecting the serum, which was then inactivated for 30 min at 56 °C, sterilized with 0.22 μm filters, and kept at −80 °C (Xu et al., 2020). Before cell treatment, the serum was diluted to 15% using a serum-free medium.

2.4. Cell lines and cultures

Zhong Qiao Xin Zhou Biotechnology Co., Ltd., Shanghai, China, provided the human hepatoma cell lines SMMC7721 and HepG2 and the normal human hepatocyte LO2 line. Cells were grown at 37 °C in a humidified incubator with 5% CO_2 in DMEM (HepG2) or RPMI-1640 medium (SMMC7721 and LO2) with 10% fetal bovine serum, penicillin (100 U/mL) and streptomycin (100 μg/mL) (Tian et al., 2015).

2.5. Microscopy

Transmission electron microscopy (TEM) and laser confocal microscopy were utilized to investigate autophagy in SMMC7721 cells. After being exposed to XJR-M for a full day, the cells were extracted, fixed for 2 h at 4 °C using 2.5% glutaraldehyde in 0.1 M phosphate buffer pH 7.4, followed by rinsing with PBS. After that, the cells were dehydrated using increasing ethanol concentrations (50–100%) and 100% acetone. They were then embedded in epoxy resin (Embed 812, Electron Microscopy Science, Perth, Australia) and incubated for 2 h at room temperature with 1% osmium tetroxide. The samples were divided into 60–80 nm segments, and lead citrate and 2% alcoholic uranium acetate were used for staining (Ogawa et al., 2001). The specimens were examined by TEM (Hitachi HT7700; Tokyo, Japan).

The cells that were transfected with lentivirus expressing GFP-RFP-LC3 (GeneChem, Shanghai, China) and inoculated in 96-well plates (5.0×10^3/well) over an overnight period were assessed through a confocal quantitative imaging cytometer (Yokogawa, Tokyo, Japan) (Yu et al., 2022; Ke et al., 2024). Cells were treated with XJR-M for 24 h, followed by a PBS wash. Each group consisted of fifty randomly chosen cells, from which the quantity of red (GFP− RFP+, autolysosomes) and yellow (GFP + RFP+, autophagosomes) puncta in individual cells was counted and averaged.

2.6. Western blotting

Cells and tumor tissues were lysed with chilled RIPA buffer and the protein concentrations were determined with a BCA Protein Assay Kit, according to the provided directions (Pérez-Revuelta et al., 2014). Forty microgram samples of protein were run on SDS-PAGE gels and electroblotted to PVDF membranes, which were blocked for 2 h with 5% BSA and treated (overnight, 4 °C) with primary antibodies. This was followed by treatment with HRP-conjugated secondary antibodies (goat anti-rabbit/anti-mouse) and visualization with a ChemiDoc MP imaging system (Bio-Rad, Hercules, CA, USA) and analysis with Image Lab software.

2.7. Gene silencing by siRNA

With certain adjustments, siRNA gene silencing was conducted, as described (Ji et al., 2019). The siRNA sequences for Beclin1 and AKT were CGACAGCAGCTGGAGCTGGATGATGAGCT and GCCAAGGAGATCATGCAGCATCGCTTCTT, respectively.

2.8. Cell viability assays

Cell viabilities after the various treatments were assessed by CCK-8 assays (Chen et al., 2024). Primarily, 5.0×10^3 cells/well were inoculated in 96-well plates until 75% confluent, and then they were subjected to 12, 24, and 36 h of diluted serum treatment. Following a 1-h incubation period at 37 °C with 10% CCK-8 reagent per well, absorbances at 450 nm were read in a multimode microplate reader (Tecan, Mannedorf, Switzerland).

2.9. Analysis of cell apoptosis through flow cytometry

The impact of XJR on apoptosis was evaluated using the Annexin V-fluorescein isothiocyanate (FITC)/propidium iodide (PI) Cell Apoptosis Detection Kit following the manufacturer's instructions (Tian et al., 2020). In 6-well plates, 1.0×10^4 HCC cells were seeded and treated for a full day. Following cell harvesting and 5 min centrifugation at 300*g*, 500 μL of binding buffer was added to the cells, followed by 5 μL each of Annexin V-FITC and PI for 15 min at room temperature. Apoptotic cells were detected by flow cytometry (FACSC, BD Instruments, USA).

2.10. Animal studies and treatment dosage

Healthy male BALB/c nude mice (age: 5–6 weeks, beginning body weight [BW]: 18.34 ± 2.26 g) were supplied by Beijing Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China). The mice were kept in a specific pathogen-free environment with unlimited water and food availability. Mice were monitored regularly for indications of distress. Following a 7-day acclimatization period, SMMC7721 cells (2.0×10^6 in 0.2 mL serum-free medium) were administered subcutaneously into the right flank of each mouse (Ji et al., 2019). Tumor cell injection was conducted under isoflurane anesthesia, with all measures taken to mitigate discomfort. After 7 days, when tumor volumes were approximately 100 mm^3, tumor-bearing mice were randomly allocated into four groups (each including 10 subjects): control (administered equivalent quantities of saline), HCQ (XJR + HCQ), the combination of XJR, HCQ (60 mg/kg BW/day), and XJR (20 mg/kg BW/day). HCQ was frequently injected intraperitoneally after being dissolved in 0.9% NaCl. Once a day, the 2.0 g/mL XJR filtrate was gavaged and diluted in 0.9% NaCl. All mice were treated continuously for 10 days with varying concentrations of the drugs. Tumor diameter was assessed following the formula;

$$V = \frac{\text{Width}^2 \text{ x Lenght}}{2}$$

Mice were euthanized post-administration by CO_2 inhalation, followed by cervical dislocation. Tumors were surgically removed post-mortem, weighed, and prepared for additional study. The Animal Ethics Committee of the Nanjing University of Chinese Medicine authorized the study protocol (Approval Number: ACU171105).

2.11. Immunohistochemistry

Immunohistochemistry (IHC) was utilized to evaluate the effects of XJR on the levels of apoptosis and autophagy markers (cleaved caspase-3, P62, and LC3A/B) in tumor tissues (Schulz et al., 2021; Liu et al., 2024). After being sectioned into 4-μm pieces and embedded in paraffin, the tumor tissues were fixed with 4% paraformaldehyde. The pieces were waxed using dimethylbenzene. The sections were treated with primary monoclonal antibodies against cleaved caspase-3, P62, and LC3A/B for a whole night at 4 °C, followed by treatment with goat anti-rabbit HRP, evaluation under light microscopy (Olympus CX41, Japan), and quantification with Image Pro Plus software.

2.12. Statistical analysis

Results are shown as means ± standard deviations. T-tests or one-way analysis of variance with subsequent Bonferroni correction were used to evaluate inter-group differences. Two-tailed p-values <0.05 were considered statistically significant.

3. Results

3.1. Results of the network pharmacology analysis of XJR in treating HCC

Through network pharmacology analysis, 38 bioactive compounds meeting the screening criteria were identified, and 480 potential targets were predicted. By intersecting these with 8269 liver cancer targets (cutoff: relevance score >2.9), 456 therapeutic targets were obtained (Fig. 1a). Based on this data, a herb-compound-target network composed of the top 100 targets for XJR was constructed and visualized. (Fig. 1b). Quercetin, associated with 142 nodes, has the highest degree value in the network. A PPI network was built to assess interactions among the top 100 intersecting genes. Based on degree value ranking, the top 10 key targets are TP53, AKT1, IL6, EGFR, SRC, HSP90AA1, TNF, IL1B, MYC, and CASP3, all of which are related to liver cancer (Fig. 1c). A more thorough enrichment study revealed 181 KEGG pathways and 1161 GO items. Fig. 1 (d, e) displays the top 10 KEGG and GO findings. These main pathways of concern are those observed in tumors, the AGE-RAGE axis in diabetic problems, the activation of receptors in chemical carcinogenesis, prostate cancer, lipid and atherosclerosis, and animal autophagy.

3.2. XJR induces autophagy in HCC cells in vitro

Changes in SMMC7721 cells after TEM assessed XJR treatment. XJR triggered the development of autophagosomes (with double-layered membranes) or autolysosomes (Fig. 2a, arrow). Conversely, no autophagosomes or autolysosomes were found in the controls. Moreover, the plasma membranes were intact and well-defined, and the morphology of the nucleus, rough endoplasmic reticulum, and other organelles was unchanged.

To assess the influence of XJR on autophagy stimulation, the levels of the autophagy-related proteins P62, Beclin1, and LC3A/B were evaluated by western blotting. In SMMC7721 and HepG2 cells, XJR significantly increased the Beclin1 and LC3A/B-II protein levels while decreasing P62 levels dose-dependent (Fig. 2b–d). Furthermore, in these cells, XJR downregulated P62 and time-dependently raised the levels of Beclin1 and LC3A/B-II (Fig. 2f–h). Fig. 2 (c, e) and S1 (a, b) display the semi-quantitative assessments. These findings showed that XJR stimulated autophagy in HepG2 and SMMC7721 cells.

Using laser confocal imaging, autophagosomes and autolysosomes in SMMC7721 cells infected with a lentivirus expressing the GFP-RFP-LC3 fusion gene were evaluated. The yellow puncta (G + R+) and red (G– R+) are autophagosomes and autolysosomes, respectively. Yellow and red puncta were significantly enhanced in HCC cells by XJR (Fig. 2j and k), indicating that XJR causes autophagy in these cells.

The autophagy inhibitor HCQ was used to test how XJR affected the autophagic flow in HCC cells. A lysosomotropic drug called HCQ prevents autophagosomes from fusing with lysosomes by inhibiting lysosomal enzymes, which causes LC3A/B-II to accumulate (Tompkins and Thorburn, 2019). In HCC cells, HCQ decreased XJR's influence on Beclin1 expression while amplifying its effect on LC3A/B-II expression (Fig. 3a and b). Western blotting was utilized for analyzing the levels of autophagy-associated proteins in transfected HCC cells after the Beclin1 gene was knocked down using siRNA technology. These findings demonstrated that Beclin1 silencing decreased or reversed the influence of XJR on LC3A/B-II expression, together with XJR-induced downregulation of P62 and overexpression of Beclin1 in HCC cells (Fig. 3c and d). These findings show that in HCC cells, XJR stimulates autophagic flux and autophagy.

3.3. In vitro, XJR induces autophagy in HCC cells via blocking the AKT/mTOR axis

The impact of XJR on the levels of proteins associated with the AKT/mTOR axis was assessed using Western blot analysis. AKT or mTOR levels were not significantly affected by XJR, while p-AKT, p-mTOR, p-p70S6K, and p-4EBP1 levels were decreased in a concentration-dependent manner (Fig. 4a and b).

To investigate the targeting of this pathway by XJR, SMMC7721 cells were treated with an AKT activator (SC), an AKT inhibitor (MK), or siRNA against AKT in the presence and absence of XJR. SC attenuated the effect of XJR on LC3A/B-II (Fig. 4c). Consequently, MK and si-AKT amplified the impact of XJR on LC3A/B-II protein levels (Fig. 4d and e). This suggests that XJR inhibits the AKT/mTOR axis to increase autophagy in HCC cells.

3.4. XJR inhibits cell viability by stimulating autophagy in vitro

The influence of XJR on the feasibility of the LO2 hepatocyte cell line was assessed. According to the findings, XJR did not affect LO2 cell viability (Fig. 5a), while that of SMMC7721 and HepG2 cells decreased with increasing XJR dosage, but not over time (Fig. 5b and c). These results indicated that XJR strongly inhibited HCC cell viability, while the viability of normal cells was unaffected.

We further explored the relationship between XJR-induced autophagy and anti-tumor activity by treating HCC cells with XJR in the presence and absence of HCQ and Beclin1 siRNA. The outcomes demonstrated that HCQ treatment and Beclin1 silencing attenuated the considerable reduction in SMMC7721 cell viability caused by XJR treatment (Fig. 5d and e). In HepG2 cells, the interventions with HCQ and Beclin1 siRNA did not significantly alter the inhibitory effect of XJR (Fig. 5f and g). This lack of a significant difference might be due to the lower sensitivity of HepG2 cells to HCQ or the involvement of alternative pathways.Thus, SMM7721 cells were selected for further analyses *in vivo*.

The levels of apoptosis-associated proteins were examined by Western blotting while flow cytometry was utilized to examine HCC cell apoptosis. This showed that XJR promoted apoptosis in SMMC7721 cells, which was reduced by HCQ and Beclin1 silencing (Fig. 6a and b). Western blotting study revealed that in SMMC7721 and HepG2 cells, XJR treatment raised the Bax, cleaved caspase-3, and lowered Bcl-2 levels; these effects were countered by HCQ treatment and Beclin1 silencing (Fig. 6c–f, Fig. S2). These studies revealed that XJR reduced HCC cell viability and caused apoptosis *in vitro*.

3.5. XJR inhibits tumor growth by stimulating autophagy in vivo

XJR's effect on tumor growth *in vivo* was assessed using an HCC xenograft mice model. The insignificant impact of XJR on body weight confirmed its minimal adverse effects on mice (Fig. 7a). Conversely, XJR time-dependently reduced tumor growth, with a 45.03% decrease in tumor weight ($p < 0.001$) (Fig. 7b and c). Furthermore, HCQ decreased the effectiveness of XJR while having no discernible influence on tumor growth (Fig. 7b and c).

The IHC findings demonstrated that XJR elevated the cleaved caspase-3 expression level in tumors. Moreover, HCQ lessened this effect (Fig. 7d). According to Western blot analysis, XJR reduced the Bcl-2

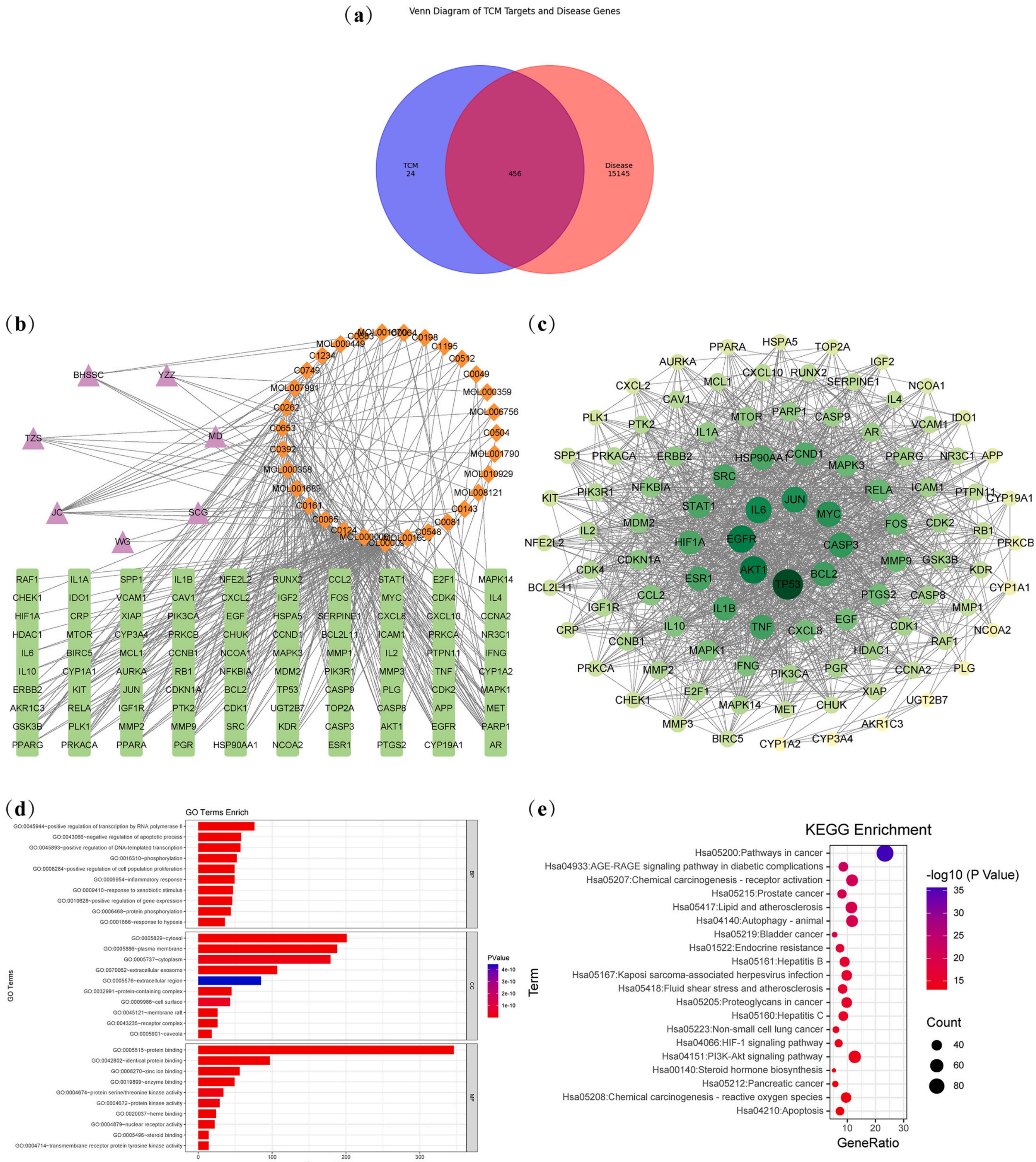

Fig. 1. Network Pharmacology diagram of XJR for treating liver cancer. **(a)** Venn diagram showing the liver cancer targets for XJR. **(b)** A herb-compound-target network for XJR, where purple denotes the 7 herbs within XJR, orange represents the compounds, and green signifies the key targets. **(c)** The top 100 core target gene maps from the PPI network. The interaction gradually increases from grass green to dark green and the size of the dots indicates the magnitude of the degree values (combined score). **(d)** The top 10 enriched entries for each GO category (BP, CC, and MF). The ordinate represents each GO term and is sorted according to the P-value; the horizontal coordinate represents the number of genes involved, the color of the column represents the P-value, the bluer, the higher the confidence.**(e)** Top 10 KEGG pathway enrichment analysis of XJR's target in treating HCC. The horizontal coordinate represents the gene ratio, while the vertical coordinate shows the term enriched by KEGG. P-values are used to order the bubble chart from smallest to largest. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

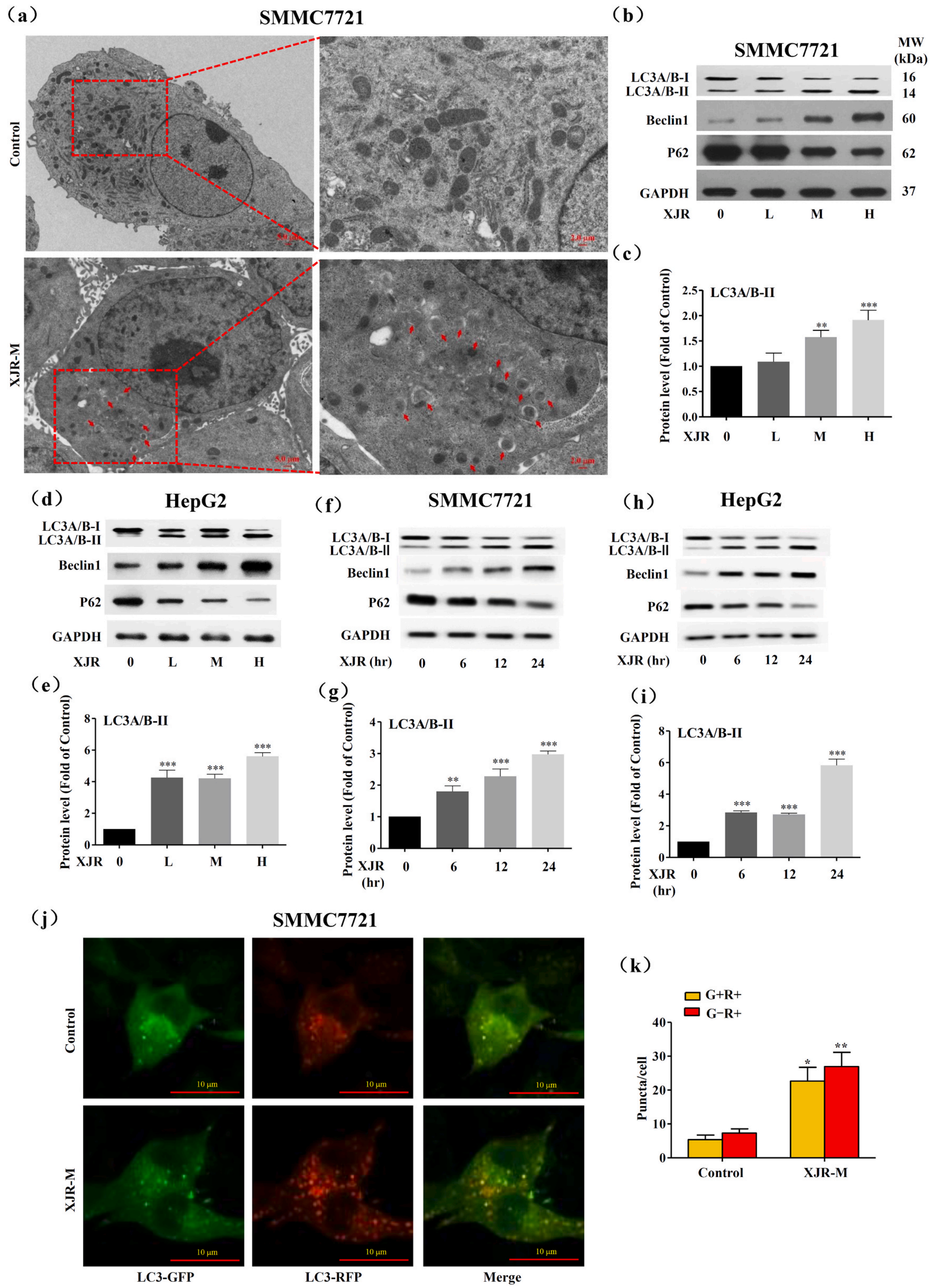

Fig. 2. Effect of XJR on autophagy in hepatocellular carcinoma cells. **(a)** Transmission electron microscopy detected autophagosomes and autolysosomes (arrows) in SMMC7721 cells after a 24-h incubation period with XJR-M. **(b, d)** Following a 24-h incubation period with XJR at varying concentrations for SMMC7721 and HepG2 cells, LC3A/B, Beclin 1, and P62 protein levels were assessed via western blotting. **(c, e)** Densitometric analysis of LC3A/B levels that correspond. **(f, h)**. HepG2 and SMMC7721 cells were treated with XJR-M for varying lengths of time. (g, i) Densitometric examination of LC3A/B-II levels that correspond. **(j-k)** GFP-RFP-LC3-expressing lentivirus was used to transfect SMMC7721 cells, and the cells were then cultured with XJR-M for a whole day. Autophagy was examined using confocal microscopy by counting the red and yellow puncta, representing autophagolysosomes and yellow puncta. XJR-M: serum from mice given a 3.78 g/kg moderate injection of XJR. Data are means ± standard deviations of three independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001 (vs. controls). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

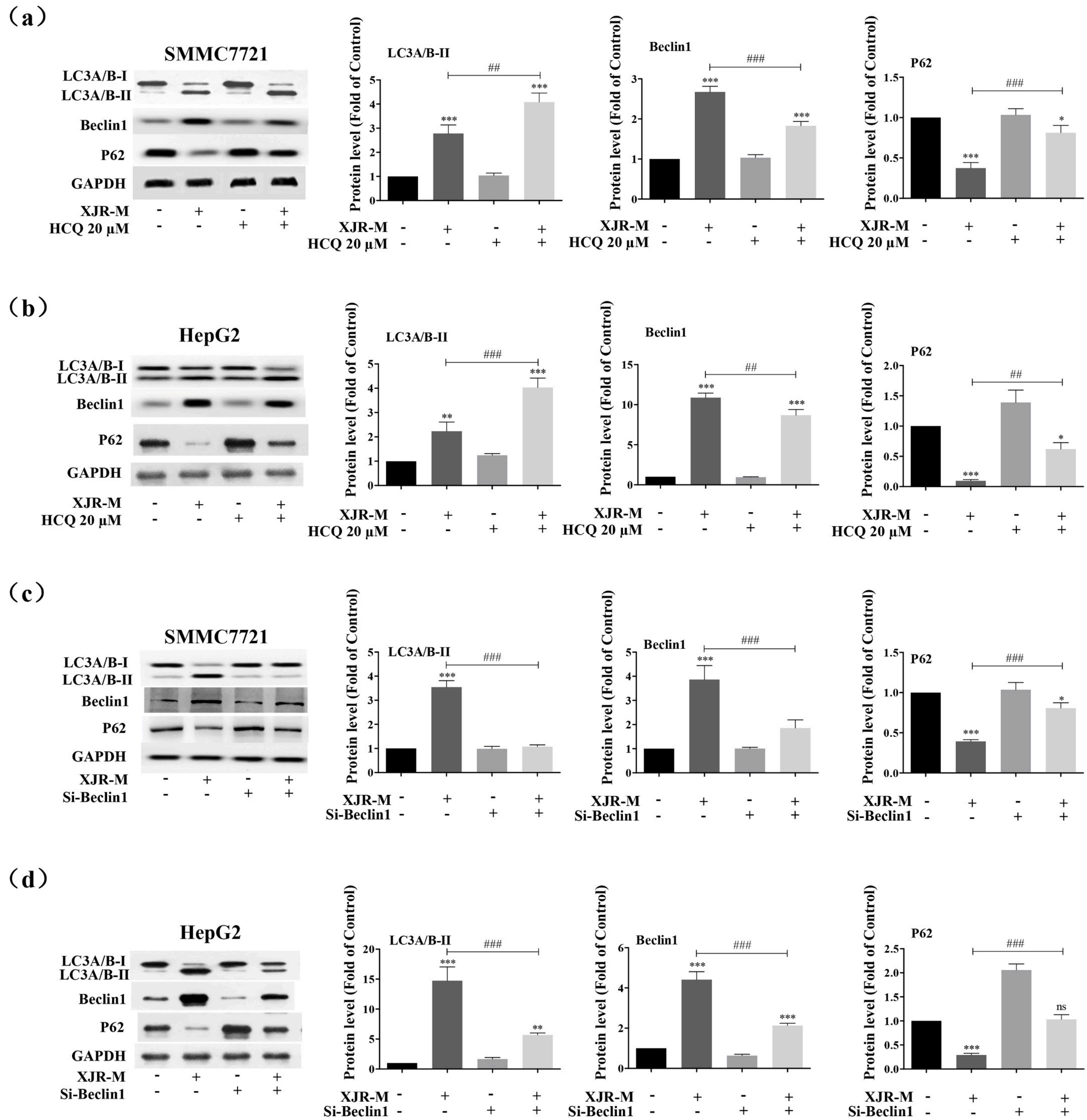

Fig. 3. In hepatocellular carcinoma cells, the effect of XJR on autophagy activation was counteracted by HCQ and Beclin 1 siRNA. **(a, b)** HepG2 and SMMC7721 cells were exposed to HCQ (20 μM) for 2 h, followed by a 24-h incubation period with XJR-M. The protein levels of P62, Beclin 1, and LC3A/B were found using western blotting. **(c, d)** HepG2 and SMMC7721 cells were treated with XJR-M for 24 h after being transfected with Beclin 1 siRNA (si-Beclin 1) or control siRNA for 48 h LC3A/B, Beclin 1, and P62 levels were quantified using western blotting. XJR-M: Mice serum was administered at a moderate XJR dosage (3.78 g/kg).

content of tumor tissues while increasing that of Bax and cleaving caspase-3. In contrast, HCQ reduced these effects (Fig. 7e). These results collectively suggested that XJR promoted tumor apoptosis by inducing autophagy.

3.6. In vivo, XJR inhibits the AKT/mTOR axis to cause autophagy

IHC indicated that XJR elevated the protein levels of LC3 while diminishing the protein levels of P62 (Fig. 8a). Moreover, western blots demonstrated that XJR markedly raised the levels of LC3A/LC3B-II and Beclin 1 while reducing those of P62 in tumor tissues (Fig. 8b). The findings indicated that XJR stimulated autophagy in a xenograft mice model of HCC.

The molecular basis of XJR-induced autophagy was studied using a tumor xenograft model. In line with the cellular findings, XJR substantially lowered the p-AKT, p-mTOR, p-p70S6K, and p-4EBP1 levels but had no discernible impact on those of AKT and mTOR (Fig. 8c). These findings imply that XJR inhibits AKT/mTOR signaling, which in turn

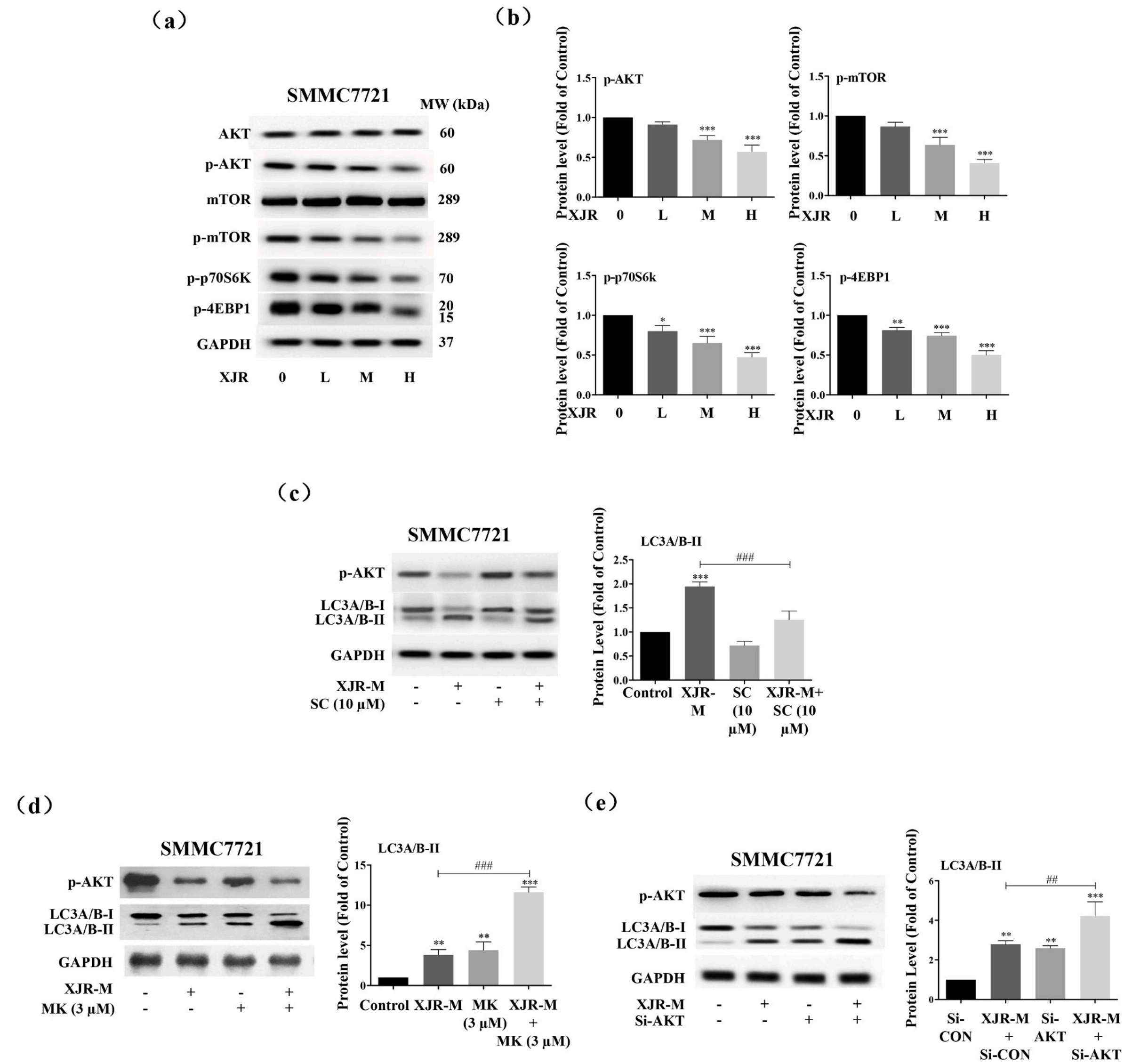

Fig. 4. Through blocking the AKT/mTOR signaling pathway, XJR promotes autophagy. **(a)** The effects of XJR on the protein levels of AKT, mTOR, p-AKT, p-mTOR, p-p70S6K, and p-4EBP1 in SMMC7721 cells treated with varying XJR concentrations during 24 h were examined using Western blot analysis. **(b)** Corresponding densitometric analyses. **(c–d)** After 2 h of treatment with SC (10 μM) or MK (3 μM), SMMC7721 cells were incubated with XJR-M for 24 h. The LC3A/B and p-AKT protein levels were determined using western blotting. **(e)**. XJR-M was incubated with SMMC7721 cells for 24 h after they were transfected with control siRNA or AKT siRNA (si-AKT) for 48 h. The protein levels of p-AKT and LC3A/B were determined using western blotting. XJR-M: serum from mice given a 3.78 g/kg moderate injection of XJR.

induces autophagy in a mouse HCC model.

4. Discussion

HCC, the main type of primary liver cancer (Xie et al., 2023), has been widely studied in terms of the function of autophagy in its onset and progression (Cui et al., 2020). XJR is a traditional formula developed by Professor Zhou Zhongying for treating malignancies. It has demonstrated encouraging therapeutic results during its many years of clinical practice use. In our previous research, we conducted several exploratory studies on the mechanisms of XJR (Li et al., 2023; Sun et al., 2023). The current study examined the mechanisms of XJR in treating HCC using network pharmacology and cellular and animal models, emphasizing its impacts on the autophagy process.

Network pharmacology results revealed potential links between several active components of the XJR formula and key regulatory targets involved in the autophagy process. Notably, the current study discovered a robust association between quercetin and the gene AKT1, which is associated with autophagy and is essential for controlling autophagy in liver cancer cells. Quercetin, a flavonoid found in many natural dietary sources, has documented antitumor properties (Tang et al., 2020). These include suppressing the function of transcription factors like Twist and controlling the expression and post-translational modification of the P53 gene (Hasan et al., 2022). Quercetin has been shown in earlier studies to promote the production of autolysosomes and autophagosomes in HCC cells. By blocking AKT/mTOR signaling and stimulating that of MAPK, it promotes autophagy, which stops HCC cell development and causes apoptosis (Ji et al., 2019).

Further analysis using GO and KEGG indicated that the pathways associated with the effects of XJR were also enriched in the autophagy process. According to (Rakesh et al., 2022), autophagy is a degradative process that can either promote or hinder the growth of tumors. It is involved in cancer. Its effects are dependent on tumor staging, specific oncogenic mutations, and the cellular environment (Debnath et al., 2023). In HCC, the activation of autophagy can lead to both apoptosis and reduced tumor growth (Kouroumalis et al., 2021, 2023). Research

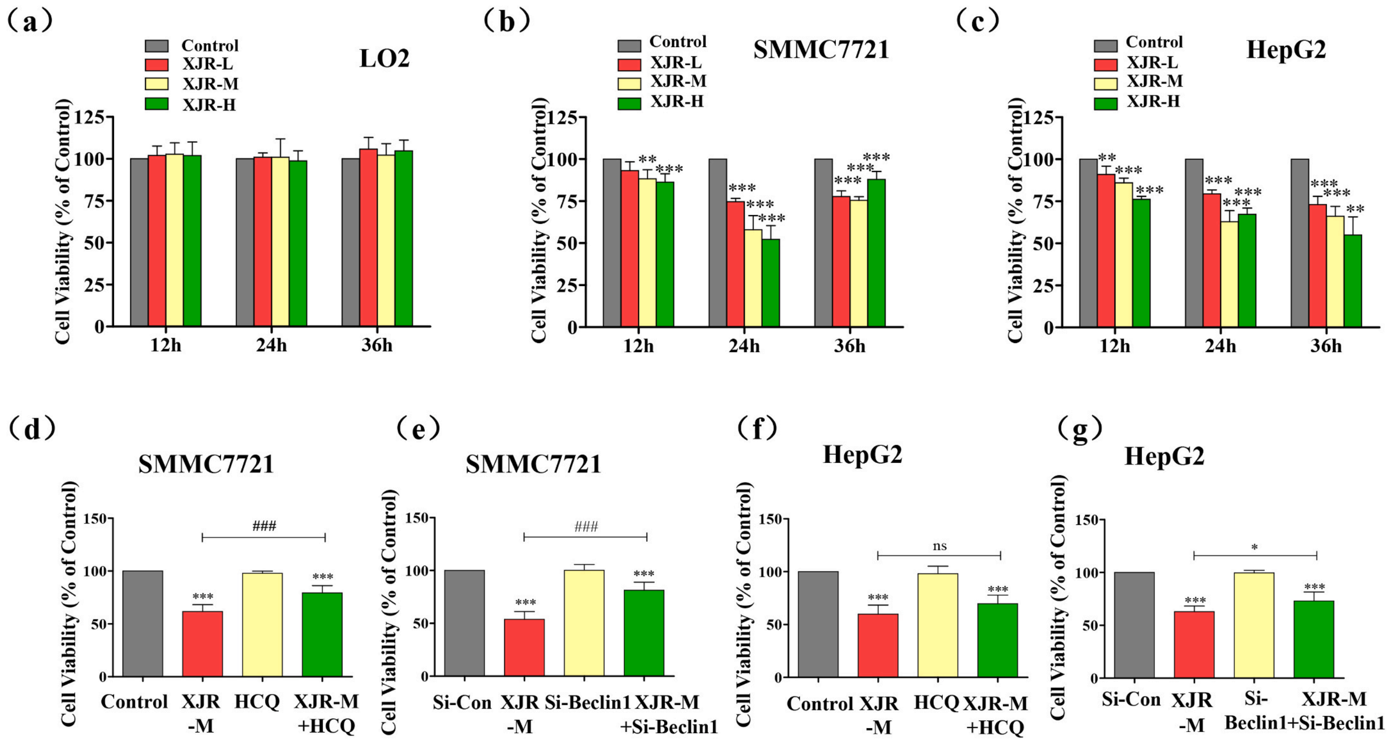

Fig. 5. *In vitro* effects of XJR on the viability of HCC. **(a–c)**. The viability of HepG2, LO2, and SMMC7721 cells that were treated with varying concentrations of XJR. **(d, f)** HepG2 and SMMC7721 cells were incubated with XJR-M for 24 h after being treated with HCQ (20 μM) for 2 h. **(e, g)** XJR-M was incubated with SMMC7721 and HepG2 cells for 24 h after transfecting with Beclin 1 siRNA (si-Beclin 1) or control siRNA for 48 h. The Cell Counting Kit-8 assay was employed to evaluate cell viability. XJR-M: serum of mice injected with a moderate dose of XJR (3.78 g/kg).

has demonstrated that several essential molecules and pathways, such as the PI3K/Akt/mTOR axis and autophagy-related factors, are the primary regulators of the autophagy process (Yang et al., 2018). PI3KCI stimulates AKT, which is further phosphorylated and prevents the TSC1/2 heterodimer from negatively regulating Rheb. This activation of mTORC1 inhibits autophagy and promotes cell growth. Additionally, AKT is activated by mTORC2, which further suppresses autophagy (Di Fazio and Matrood, 2018; Peng et al., 2022). A majority of patients with HCC show activation of the PI3K/AKT/mTOR axis, a finding that provides a crucial basis for developing targeted therapeutic strategies (Tian et al., 2023). As a result, scientists have created anti-HCC medications targeting this signaling system, and clinical trials have confirmed their efficacy (Patra et al., 2020; Yan et al., 2022). Therefore, building on our previous findings, we aim to explore whether XJR also affects the AKT/mTOR pathway through autophagy. Our results have demonstrated that XJR reduces cell survival and promotes apoptosis in HCC cell lines by activating autophagy. Furthermore, XJR promotes autophagy by inhibiting the AKT/mTOR axis.

Similarly, studies in the mouse models demonstrated that XJR activated autophagy via the AKT pathway. These findings are supported by the fact that autophagy suppression lessened the influence of XJR on tumor growth and apoptosis. These results offer solid experimental support for the notion that XJR suppresses HCC growth by promoting autophagy.

In the present investigation, we employed multiple techniques to evaluate the regulatory impact of XJR on autophagy, including morphological inspection and quantification of autophagy marker protein expression levels. In addition, we combined chemical inhibitors and siRNA knockdown techniques to demonstrate further. Our more comprehensive study uses both *in vitro* cells and transplanted tumor models. This investigation assessed associations between autophagy and apoptosis as well as between autophagy and cell activity. However, the molecular processes underlying these associations were not evaluated, and the sample size in each subgroup of *in vivo* experiments was limited. Cellular autophagy is associated with both cellular survival and quality control in eukaryotic cells and is crucial for maintaining cellular homeostasis (Hernandez and Perera, 2022). The increased understanding of autophagy assists the development of novel treatments, especially for cancer.

Moreover, controlling autophagy provides a novel approach to cancer management and therapy (Rangel et al., 2022; Salimi-Jeda et al., 2022). Numerous studies have assessed TCM's impact on autophagy, including its function in advancing cancer and potential therapeutic uses. Artesunate reduces the invasive ability of cisplatin-resistant bladder cancer cells by targeting autophagy (Zhao et al., 2020). Some herbal formulations can regulate autophagy in tumor cells. Moreover, the TCM formulation T33 decreases human colorectal cancer cell activity by inducing autophagy in both cellular and animal models (Liu et al., 2022). The anti-tumor efficacy of XJR remains a significant concern, especially its role in autophagy, despite its proven anti-tumor effects in certain cancers (Y. Li et al., 2020; Qiu et al., 2020; Wang et al., 2019).

The present research evaluated the levels of the autophagy-associated proteins LC3A/B, Beclin1, and P62 to elucidate the effects of XJR on autophagy. An Atg8 homolog known as microtubule-associated protein 1 light chain 3 (LC3) strongly correlates with autophagosomes (Klionsky et al., 2021). To develop autophagosomes, cytosolic LC3-I must transform into membrane-bound LC3-II (Klionsky et al., 2021). LC3 has three isoforms (A, B, and C) in mammals, and LC3-B is a well-known autophagy marker. LC3-C reportedly does not contribute to cytosolic autophagy (Baeken et al., 2020). Therefore, antibodies against LC3-A and LC3-B were used in this study. Beclin1, a marker of autophagy initiation, has been used to monitor autophagy (Klionsky et al., 2021). The P62 protein, an autophagy receptor, directly binds to LC3 and undergoes autophagic degradation (Aparicio et al., 2020). TEM and laser confocal microscopy were also used to evaluate

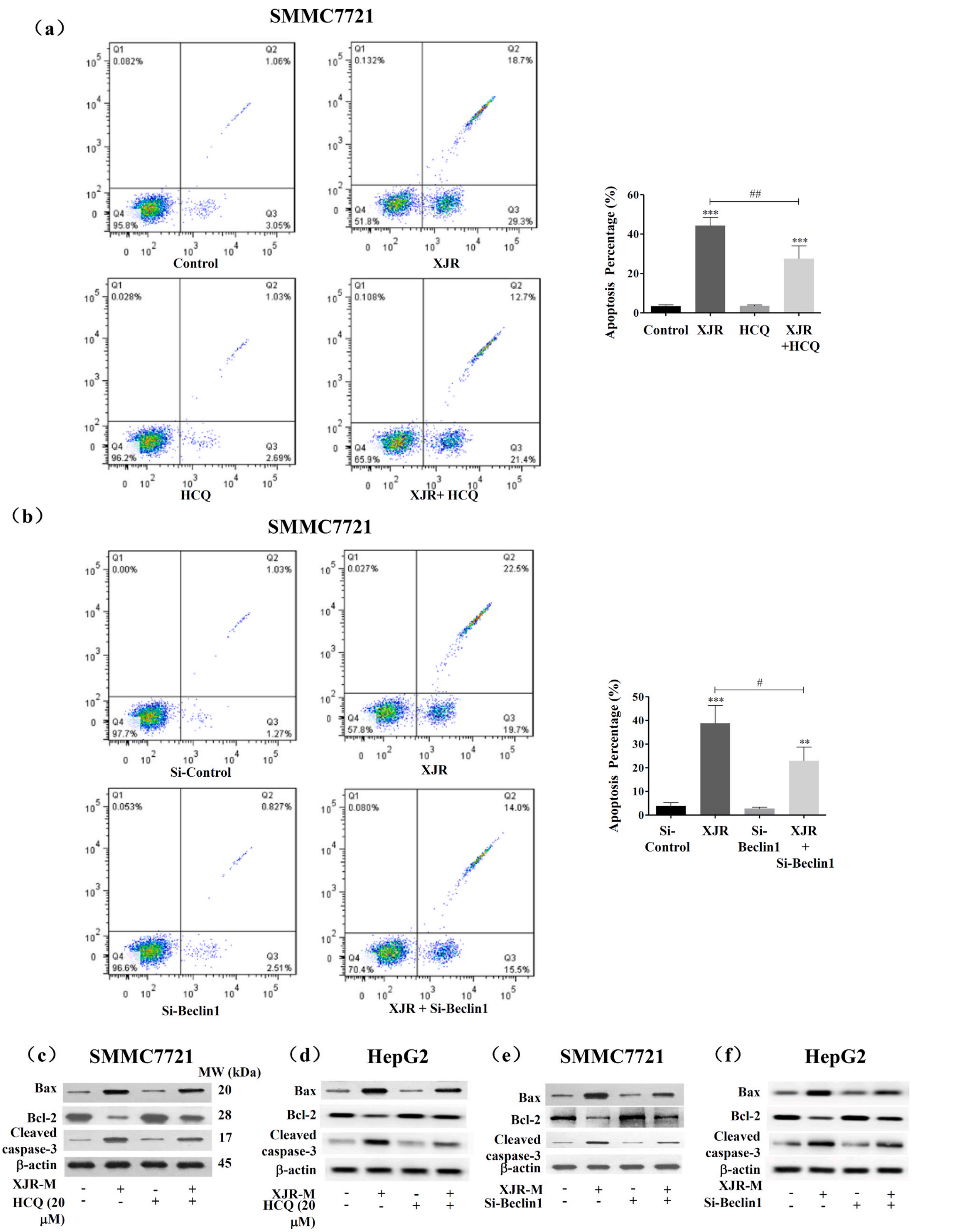

Fig. 6. Impact of XJR on *in vitro* apoptosis. **(a–b)** SMMC7721 and HepG2 cells were exposed to HCQ (20 μM) for 2 h, followed by a 24 h incubation period with XJR-M. Flow cytometry was used to identify apoptosis. **(c–d)** HepG2 and SMMC7721 cells were exposed to HCQ for 2 h, and then XJR-M was added for 24 h **(e-f)** HepG2 and SMMC7721 cells were treated with XJR-M for 24 h after transfecting them with Beclin 1 siRNA (si-Beclin 1) or control siRNA (si-CON) for 48 h. The protein levels of Bax, Bcl-2, and cleaved caspase-3 were determined by western blotting. Three separate independent runs of the experiments were conducted. In Fig. S2, densitometry analyses are displayed. XJR-M: serum of mice injected with a moderate dose of XJR (3.78 g/kg).

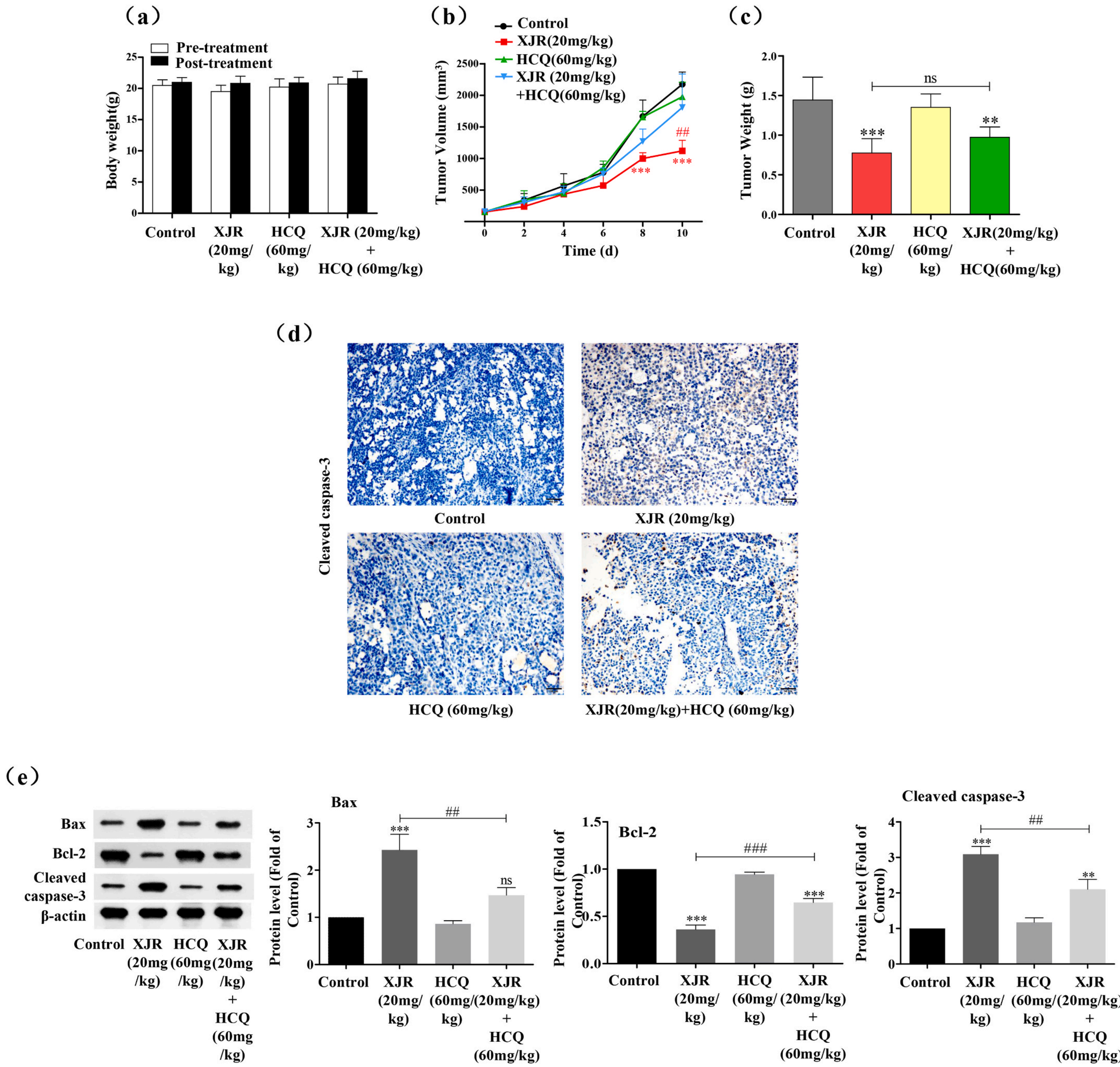

Fig. 7. Impact of XJR on the *in vivo* development of SMMC7721 tumors. **(a–c)** Body weight, tumor weight, and tumor volume in mice harboring tumors treated with HCQ (60 mg/kg), XJR (20 mg/kg), XJR + HCQ, or control. **(d)** The protein expression of cleaved caspase-3 in tumor tissue was examined using immunohistochemistry. **(e)** Western blot analysis determined the protein levels of cleaved caspase-3, Bcl-2, and Bax in tumor tissue. On the right, densitometric analyses are displayed.

autophagy and the contents of autophagy markers in untreated and silenced cells, as well as after treatment with autophagy inhibitors. The results showed that XJR activated autophagy in HCC cells. Later research found that XJR induced autophagy, which reduced cell viability and caused death in HCC cells *in vitro*—possibly via blocking AKT/mTOR. Similarly, autophagy in HCC xenograft cells was triggered by XJR via the AKT/mTOR pathway in animal experiments. In line with these results, XJR's effects on tumor growth and apoptosis were mitigated by autophagy suppression.

Both mTOR-dependent and independent mechanisms regulate autophagy. However, autophagy is regulated primarily by the PI3K/AKT/mTOR axis (Al-Bari and Xu, 2020), and TCM used in cancer therapies regulates autophagy by targeting this pathway. For example, Alpinia katsumadai Hayata activates autophagy via the AKT/mTOR/p70S6K axis, causing apoptosis in human cancer cells (An et al., 2022). Through the AKT/mTOR axis, Astragalus-Scorpion activates autophagy to prevent prostate cancer (You et al., 2022). Examining the AKT/mTOR-associated components expressed in HCC cells, we found that XJR decreased both p-AKT and p-mTOR levels. Moreover, the results of assays using AKT activators and inhibitors and siRNA-based knockdown confirmed that XJR induced autophagy in HCC cells through this pathway.

XJR is a traditional Chinese medicine (TCM) formula that has been widely used in clinical practice. Both clinical observations and our current study suggest that XJR not only exhibits promising anti-tumor efficacy but may also have beneficial effects on the immune system

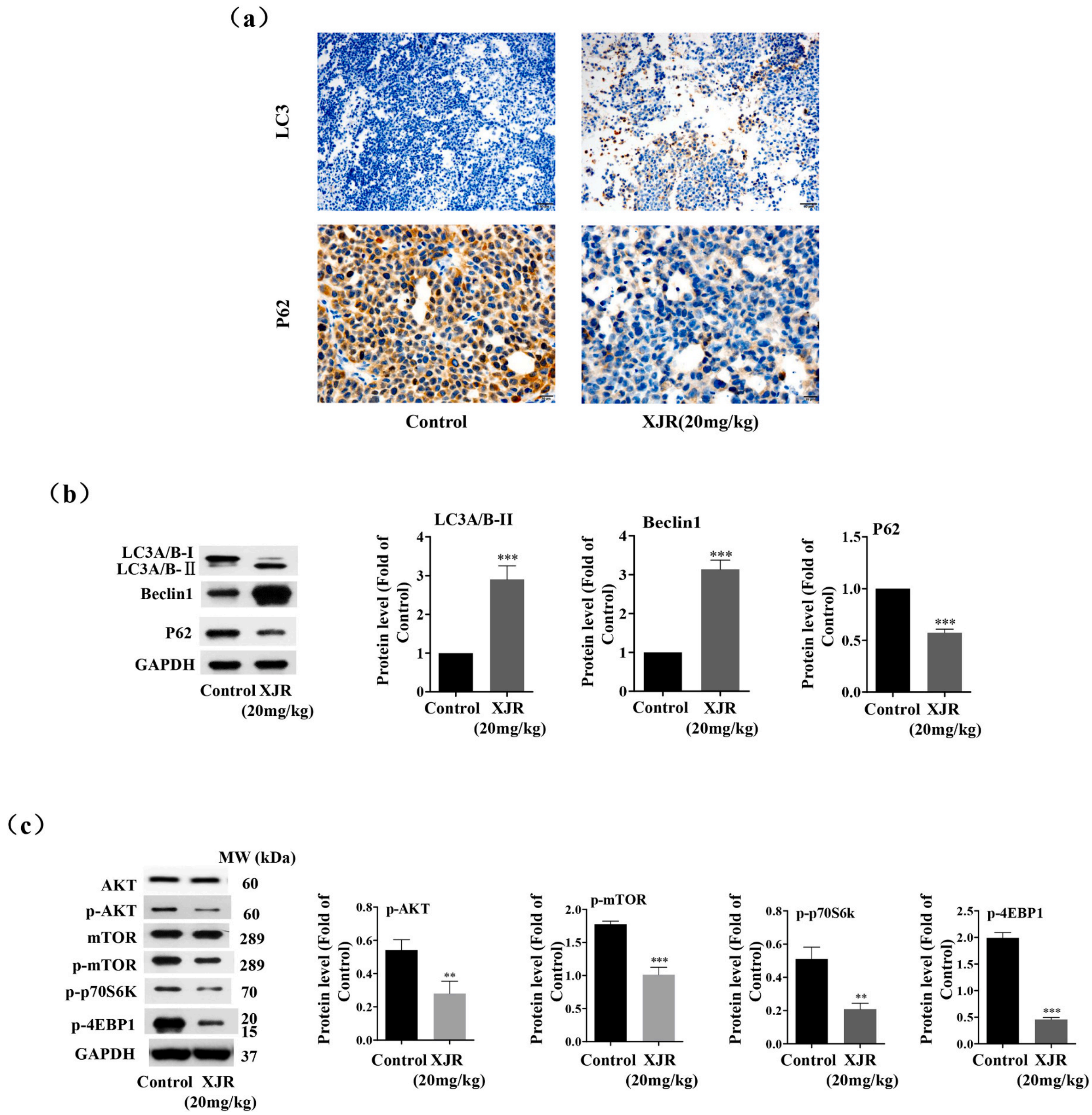

Fig. 8. Through its *in vivo* inhibition of the AKT/mTOR signaling pathway, XJR promotes autophagy in SMMC7721 cells. **(a)**. The protein expression of the autophagy markers P62 and LC3A/B in tumor tissue was examined using immunohistochemistry. **(b)**. Beclin 1, P62, and LC3A/B-I, II protein levels in tumor tissue were analyzed using Western blot. On the right are densitometric analyses. **(c)** protein levels of p-4EBP1, p-p70S6K, p-mTOR, mTOR, p-AKT, and AKT in tumor tissue were examined by Western blot analysis. On the right are densitometric analyses.

and other physiological aspects. While A et al. have reported that Ginsenoside Rk3 inhibits hepatocellular carcinoma (HCC) growth by inducing autophagy and apoptosis via the PI3K/AKT pathway (Qu et al., 2023), XJR presents a competitive advantage due to its multi-faceted mechanisms and pathways of action compared to such single-compound approaches.

Moreover, although there are reports of other TCM ingredients, such as cantharidin (Yan et al., 2023) and *Scutellaria barbata* D.Don (Yang et al., 2021), inducing autophagy in HCC, XJR, as a multi-component formula, offers unique benefits. Certain components may directly inhibit tumor cell growth, while others may indirectly enhance anti-tumor efficacy by boosting the immune response or improving the tumor microenvironment. This synergistic effect not only allows for lower dosages but also enhances overall therapeutic efficacy and reduces potential side effects.These findings clarify the anti-cancer properties of Traditional Chinese Medicine and illustrate its potential use in cancer treatment.

5. Conclusion

Our findings provide robust experimental evidence that XJR inhibits HCC through the stimulation of autophagy. As shown in both cellular and animal models, XJR activated autophagy by blocking the AKT/mTOR axis, which caused apoptosis and decreased cell survival in HCC cells. Furthermore, XJR showed few adverse effects on normal cells, supporting its therapeutic use in HCC.

CRediT authorship contribution statement

Yi Ji: Methodology, Investigation. **Li Li:** Methodology, Investigation. **Wenting Li:** Methodology. **Liu Li:** Writing – original draft. **Yanxia Ma:** Writing – review & editing. **Qingfeng Li:** Writing – review & editing. **Xi Chen:** Data curation. **Wenyue Zhao:** Data curation. **Hengzhou Zhu:** Funding acquisition, Data curation. **Jiege Huo:** Conceptualization. **Mianhua Wu:** Funding acquisition, Conceptualization.

Ethics approval and consent to participate

The study protocol was approved by the Animal Ethics Committee of Nanjing University of Chinese Medicine (Approval Number: ACU171105).

Consent for publication

The authors declare that they have no competing interests.

Availability of data and materials

Not applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

BW	body weight
CCK-8	Cell Counting Kit-8
DL	Drug-Likeness
FITC	fluorescein isothiocyanate
GO	Gene Ontology
HCC	hepatocellular carcinoma
HRP	horseradish peroxidase
KEGG	Kyoto Encyclopedia of Genes and Genomes
OB	Oral Bioavailability
PI	propidium iodide
PPI	Protein-Protein Interaction
TCM	traditional Chinese medicine
TEM	transmission electron microscopy
XJR	Xiaoai Jiedu recipe.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2024.119135.

Data availability

Data will be made available on request.

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Corrigendum

Corrigendum to “Xiaoai Jiedu recipe reduces cell survival and induces apoptosis in hepatocellular carcinoma by stimulating autophagy via the AKT/mTOR pathway” [J. Ethnopharmacol. 339 (2025) 119135]

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The authors regret < that a layout error in Fig. 2.(h) occurred during the process of pasting images and creating a unified layout, as we handled a large number of similar images at once. We have reviewed all the original images and confirmed that they are correct. We have also reorganized Fig. 2 using the original images to present the correct version. The WB images shown in the article’s figures were the representative images from each group, while the quantitative data in the figures were analyzed based on three sets of parallel original and correct images. Therefore, the authors confirmed that the corrections made to the images in this corrigendum do not affect the results of the study. The corrected Fig. 2 has also been provided below>.

The authors would like to apologise for any inconvenience caused.

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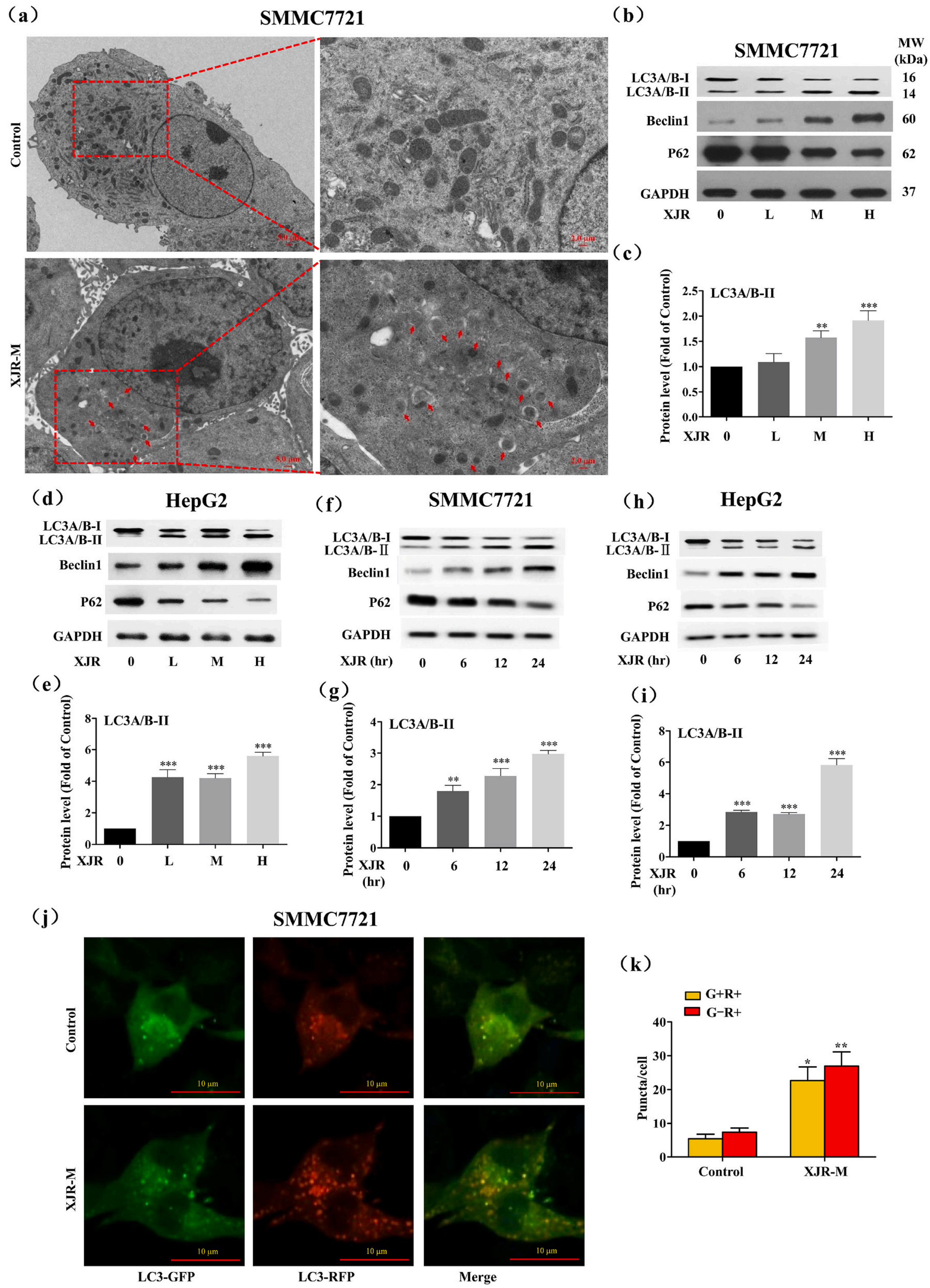

Fig. 2. Effect of XJR on autophagy in hepatocellular carcinoma cells. **(a)** Transmission electron microscopy detected autophagosomes and autolysosomes (arrows) in SMMC7721 cells after a 24-h incubation period with XJR-M. **(b, d)** Following a 24-h incubation period with XJR at varying concentrations for SMMC7721 and HepG2 cells, LC3A/B, Beclin 1, and P62 protein levels were assessed via western blotting. **(c, e)** Densitometric analysis of LC3A/B levels that correspond. **(f, h)**. HepG2 and SMMC7721 cells were treated with XJR-M for varying lengths of time. **(g, i)** Densitometric examination of LC3A/B-II levels that correspond. **(j–k)** GFP-RFP-LC3-expressing lentivirus was used to transfect SMMC7721 cells, and the cells were then cultured with XJR-M for a whole day. Autophagy was examined using confocal microscopy by counting the red and yellow puncta, representing autophagolysosomes and yellow puncta. XJR-M: serum from mice given a 3.78 g/kg moderate injection of XJR. Data are means ± standard deviations of three independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001 (vs. controls).